

# Linear algebra and matrices

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ECM3446/ECMM461: High Performance Computing

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# Module Overview

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|-------------------------------------|------------------------------------------------|---------------------------------------------------------|
| Week 1: Introduction to HPC         | Week 5: Floating point arithmetic              | Week 8: Task parallelism and manager worker parallelism |
| Week 2: Introduction to OpenMP      | Week 6: Memory and cache                       | Week 9: Linear algebra and matrices                     |
| Week 3: Numerical solutions to PDEs | Week 7: Factors affecting parallel performance | Week 10: Accelerators and GPUs                          |
| Week 4: Introduction to MPI         |                                                | Week 11: Summary                                        |

# Syllabus Plan

- Motivation of and introduction to high-performance computing
- Parallel computer architectures: shared-memory and distributed-memory architectures, multi-core processors, Graphics Processing Units (GPUs)
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

# Lecture Units

- Unit 9.1: Linear algebra in HPC
- Unit 9.2: Dense matrices and BLAS
- Unit 9.3: Sparse matrices and compressed sparse row format
- Unit 9.4: Linear algebra benchmarks

## Workshop

Investigate the performance of dense matrix-matrix multiplication on ISCA